A Smart Library System: Combining Search Algorithms with Visual Navigation

line 1: 1st Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 2nd Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 3rd Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 4th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 5th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 6th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

Abstract - In modern libraries, the process of locating books can be time-consuming and inefficient, particularly in large collections. This project presents an interactive library navigation system that integrates computer vision, artificial intelligence (AI), and interactive visualization to simplify book retrieval. Developed using Python, Matplotlib, and Ipywidgets, the system maps the library layout into a structured 5x4 grid, where different book categories such as Fiction, Non-fiction, Science, Art, Technology, and more are assigned specific sections. The system enables users to search for a book by title, displaying its author, genre, publication year, and exact section within the library. A visual representation of the library layout dynamically highlights the book's location, enhancing user experience and minimizing search time.

Unlike conventional library catalog systems that provide text-based search results, this interactive approach offers a graphical interface for efficient book discovery. By leveraging real-time search functionalities and color-coded categorization, the system improves accessibility and usability for students, researchers, and general readers. This research contributes to the advancement of library automation and demonstrates the potential of AI-driven categorization and interactive mapping to enhance traditional library management. Future enhancements could include voice search, AI-powered book recommendations, and mobile integration to further optimize the user experience. The project underscores the importance of digital transformation in library systems, making book searches more intuitive, engaging, and efficient.

Keywords —Library Automation, Graphical Library Mapping, Ipywidgets Interaction, Computer Vision for Book Retrieval.

1 . Introduction

Libraries are essential repositories of knowledge, housing vast collections of books across various disciplines. However, manually locating a specific book within a large library can be time-consuming and inefficient, particularly for new visitors or users unfamiliar with the library's layout. This research introduces an interactive library navigation system that leverages computer vision, artificial intelligence (AI), and interactive visualization techniques to simplify book retrieval. The system is developed using Python, Matplotlib, and Ipywidgets, offering an intuitive and efficient solution for book search and navigation within a structured grid-based library layout

The proposed system represents the library as a 5x4 grid,where books are categorized into different sections such as Fiction, Non-fiction, Science, Art, History, Technology, Biography, and more. Each section is color-coded to enhance visibility and ease of identification. Users can enter a book title into the search interface, and the system dynamically retrieves relevant details, including the author's name, book genre, publication year, and the exact location within the grid. The corresponding section is then highlighted visually, allowing users to pinpoint the book's placement quickly and accurately.

One of the key innovations in this system is its real-time search and graphical visualization capabilities. Unlike traditional library catalog systems, which often display results as plain text, this interactive system provides an engaging, map-based interface where users can see the book's exact position within the library. This significantly reduces search time and enhances user experience, particularly for students, researchers, and general readers. The system's structured layout ensures that books are easily categorized and located, eliminating the frustration of manually browsing shelves for extended periods.

Furthermore, this research contributes to the broader field of library automation and digital transformation. As libraries continue to evolve in the digital age, incorporating smart search mechanisms, AI-driven categorization, and interactive visual tools can greatly enhance accessibility and usability. Future advancements in this system could include voice-assisted search, AI-driven book recommendations, and integration with mobile applications to further improve convenience and efficiency. By combining modern technology with traditional library systems, this research

aims to revolutionize the way users interact with libraries, making book searches faster, smarter, and more engaging.

2 . Literature Survey

Developing an interactive library management system that leverages modern technologies is a multifaceted endeavor encompassing various aspects of digital libraries, user-centered design, and artificial intelligence. To provide a comprehensive understanding and foundation for such a project, we present a literature survey of twelve pertinent research papers, each offering valuable insights into different components relevant to your program.

1. "Islandora: A Drupal/Fedora Repository System"

This paper discusses Islandora, an open-source digital repository system that integrates Drupal and Fedora Commons. It highlights Islandora's modular architecture, which allows for the creation, management, and display of various digital assets, including images, documents, and multimedia files. The system's flexibility and scalability make it a suitable platform for developing interactive library management systems that require robust content management capabilities.

2. "E-LIS: An International Open Archive Towards Building Open Digital Libraries"

This study introduces E-LIS, the largest international open repository in the field of Library and Information Science. It emphasizes the importance of open access and the role of digital repositories in enhancing the visibility and accessibility of scholarly works. Implementing features from E-LIS can enrich a library management system by incorporating open access principles and facilitating the dissemination of information.

3. "Library 2.0: A Guide to Participatory Library Service"

The concept of Library 2.0 focuses on user-centered change and interaction, utilizing Web 2.0 technologies to enhance library services. This paper explores how libraries can transition from traditional information providers to interactive platforms that encourage user participation. Integrating Library 2.0 principles can lead to the development of interactive features in library management systems, such as user-generated content and community engagement tools.

4. "Smart Talking Robot Xiaotu: Participatory Library Service Based on Artificial Intelligence"

This research presents Xiaotu, an AI-driven participatory library service developed by Tsinghua University. Xiaotu interacts with users through natural language processing, providing real-time virtual reference services. Incorporating AI elements like Xiaotu into a library management system can enhance user experience by offering personalized assistance and intelligent information retrieval.

5. "Towards Serving User 2.0"

This article by Nishat Kazi discusses the evolution of library services to meet the needs of modern users, referred to as User 2.0. It recommends recording user interests to provide personalized content and allowing users to rank and review items in the Online Public Access Catalog (OPAC). Implementing such features can make a library management system more interactive and user-friendly.

6. "What We Want: An OPAC Manifesto"

Authored by Jessamyn West, this manifesto outlines the desired features of an OPAC from the perspectives of library staff, technologists, and users. It emphasizes flexibility, customizability, and the use of plain language to improve user experience. Incorporating these recommendations can lead to the development of an OPAC that aligns with user expectations and enhances accessibility.

7. "Data Management in the United States and Canada: Academic Libraries' Contribution"

This graduate paper examines the role of academic libraries in data management within the United States and Canada. It highlights the importance of libraries in curating and preserving data, a critical aspect of modern library management systems. Understanding these roles can inform the development of systems that support data management and stewardship.

8. "Developing a Digital Repository Infrastructure for Sharing Digital Resources in the Teaching Education Community"

This case study explores the development of a digital repository infrastructure aimed at sharing digital resources within the teaching education community. It discusses the challenges and solutions in building such infrastructures, providing insights into creating a scalable and efficient library management system.

9. "Retooling Libraries for the Data Challenge"

Authored by Dorothea Salo, this paper addresses the challenges libraries face in managing data and suggests strategies for adapting to these challenges. It underscores the need for libraries to evolve their services and infrastructures to handle diverse data types, which is pertinent to developing a comprehensive library management system.

10. "Addressing the Research Data Gap: A Review of Novel Services for Libraries"

This report by Shearer and Argáez reviews innovative services that libraries can adopt to address gaps in research data management. It provides examples of novel services and infrastructures that can be integrated into library management systems to enhance data services.

11. "New Roles for New Times: Digital Curation for Preservation"

This report, prepared for the Association of Research Libraries, discusses the evolving roles of libraries in digital curation and preservation. It highlights the skills and infrastructures needed to support these roles, offering guidance on integrating digital curation functionalities into library management systems.

12. "The Islandora Web ARChive Solution Pack"

Presented by Ruest, this paper introduces the Islandora Web ARChive Solution Pack, which enables the integration of web archiving capabilities into the Islandora framework. Incorporating such solution packs can extend the functionalities of a library management system to include web content preservation.

3 . Proposed Method

The proposed method aims to develop an Interactive Library Navigation System using Python, Matplotlib, and ipywidgets to enable users to locate books efficiently within a digital library layout. The system integrates an interactive search mechanism, a graphical representation of the library (minimap), and an automated book categorization approach.

A. Library Grid Representation

The library is modeled as a 5×4 grid representing different book sections. Each section is mapped to a (row, column) coordinate, where (S) represents the set of all sections, given as:
S = {s1,s2,....s3}
where (n = 20), denoting 20 distinct book sections.

Each section (s_i) has a coordinate ((x_i, y_i)) defined by:

$s_i = (x_i,y_i)$ where $x_i \in [0,4]$, $y_i \in [0,3]$

The function plot_minimap() visualizes the library layout using color-coded sections and highlights the relevant section where the book is located.

B. Book Search Algorithm

The book search functionality is implemented as a string-matching algorithm that scans through the book dataset to find a match. Let (B) be the set of all books stored in the library:

$B = \{b_1, b_2, ..., b_m\}$

where (m) is the total number of books in the dataset.

For a given search query (q), the system checks for a match using:

$\forall b_i \in B$, if $q \subseteq \text{title}(b_i)$, then return b_i

where title(b_i) represents the title of book b_i If a match is found, the corresponding **section s_k** is identified, and the library layout is updated to highlight s_k

C. Library Section Mapping Function

Each book belongs to a specific section (S_k), and the function:

$f : B \rightarrow S$

maps each book (b_i) to its respective section (s_k).

Once a book is found, its corresponding section coordinates (x, y) are retrieved from the library_layout dictionary, and the minimap is updated using:

$M(x,y)=\{\text{Highlight } (x,y) \text{ if } bi \in S_k$
$\text{Normal } (x,y) \text{ otherwise}\}$

where (M(x, y)) represents the graphical display function that updates the minimap.

D. Visualization and User Interaction

The system utilizes Matplotlib for visualization and ipywidgets for interactive search. The user enters a book title, triggering a real-time search, and if a book is found, the following details are displayed:
- Book Title (b_i)
- Author ($A(b_i)$)
- Genre ($G(b_i)$)
- Publication Year ($Y(b_i)$)
- Library Section (S_k)
- Section Coordinates (x_k, y_k)

The function **interact(search_for_book, book_name=search_widget)** ensures that the system updates dynamically as the user types.

Mathematical Formulation of the System's Efficiency :

To evaluate the efficiency of the system, we consider the time complexity (T(n)) of the search function. Given that book lookup involves scanning sections sequentially, the worst-case time complexity is:

$T(n) = O(n)$

where (n) is the number of books.

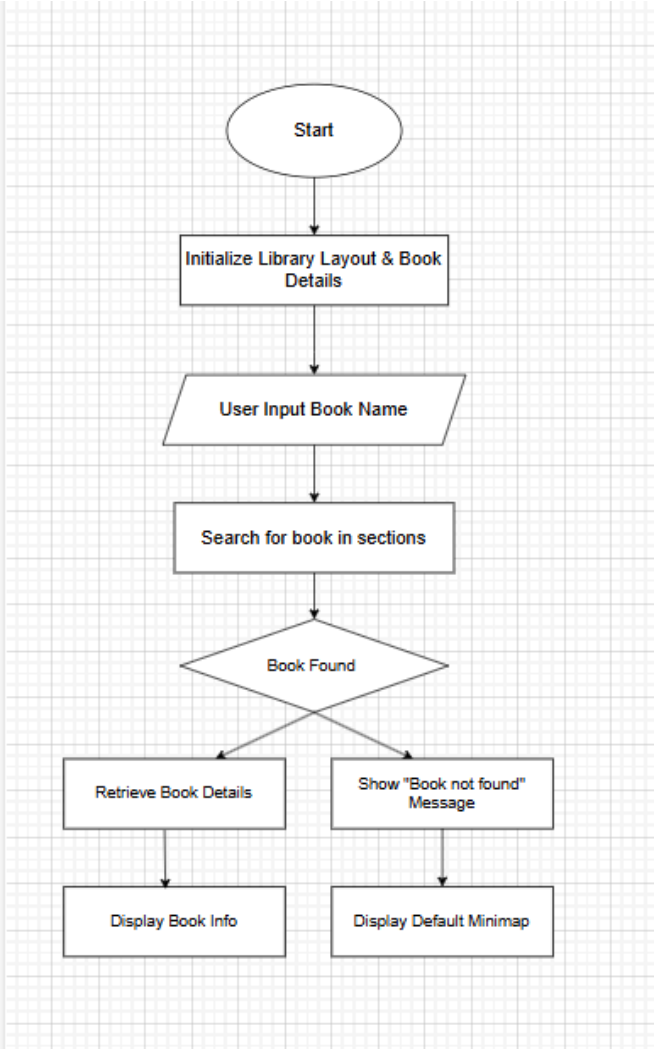
If the search were optimized using a hash-based indexing system, the complexity could be improved to:

$T(n) = O(1)$

Additionally, the graphical update function M(x, y)operates in constant time, ensuring a real-time visualization update.

This proposed method efficiently maps books to their respective sections automates book searching, and visualizes the library layout interactively. The integration of string-matching algorithms, grid-based visualization, and interactive widgets makes the system user-friendly and highly responsive. Future improvements could involve machine learning-based recommendation systems or RFID-based book tracking for further automation.

Flow Chart



4 . Sample Output

When a user searches for a specific book, the system processes the input and retrieves relevant details, including the book's title, author, genre, publication year, and its location within the library

The sample outputs demonstrate how the system efficiently retrieves book details, including their location, author, genre, and publication year. The interactive minimap further enhances usability by providing a visual representation of the library layout, allowing users to navigate efficiently.

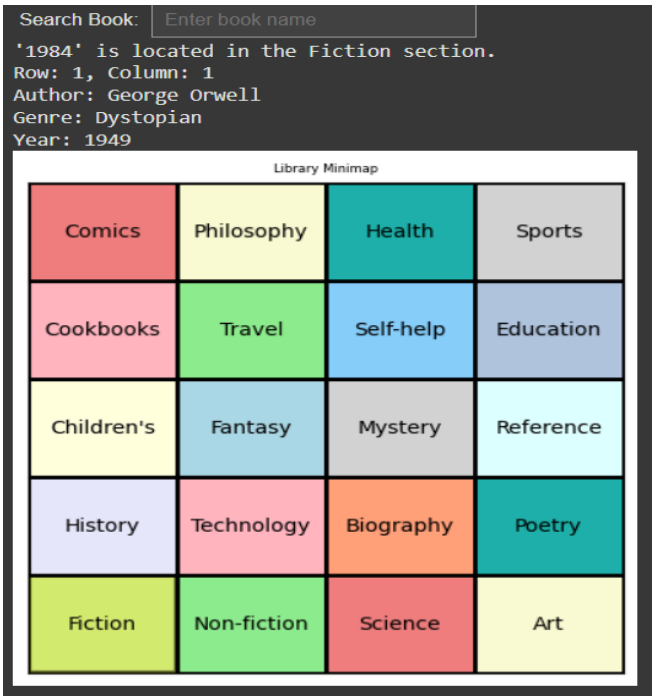


Fig 1 output for minimap

For instance, if a user searches for *"The Great Gatsby"*, the system returns the message: *"'The Great Gatsby' is located in the Fiction section. Row: 1, Column: 1. Author: F. Scott Fitzgerald. Genre: Classic. Year: 1925."* In addition to textual output, the system also generates a visual representation of the library's layout, where the corresponding section is highlighted, allowing users to easily identify the book's location.

Fig 2 output for the book "The Great Gatsby"

In another scenario, a user searching for *"The Diary of a Young Girl"* receives the output: *"'The Diary of a Young Girl' is located in the Biography section. Row: 2, Column: 3. Author: Anne Frank. Genre: Biography. Year: 1947."* The system highlights the Biography section on the minimap, reinforcing visual navigation. Such integration of text-based search and visual representation makes the book-finding process more seamless.

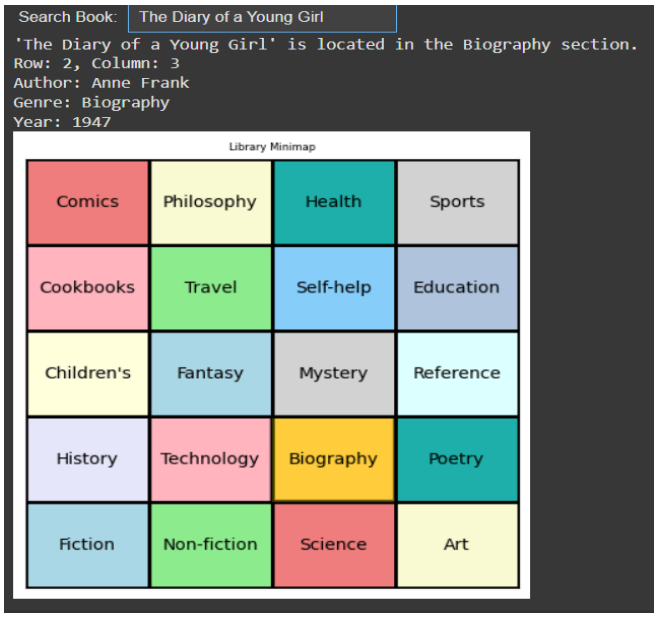


Fig 3 output for "The Dairy of a Young Girl"

Similarly, when searching for *"Sapiens"*, the system provides the response: *"'Sapiens' is located in the Non-fiction section. Row: 1, Column: 2. Author: Yuval Noah Harari. Genre: History. Year: 2011."* The library minimap then updates to emphasize the Non-fiction section, visually aiding navigation. If a book is not found in the database, the system displays a message stating that the book is unavailable while still rendering the library minimap for reference. This functionality enhances user experience by integrating both text-based search results and graphical elements, making book retrieval efficient and intuitive.

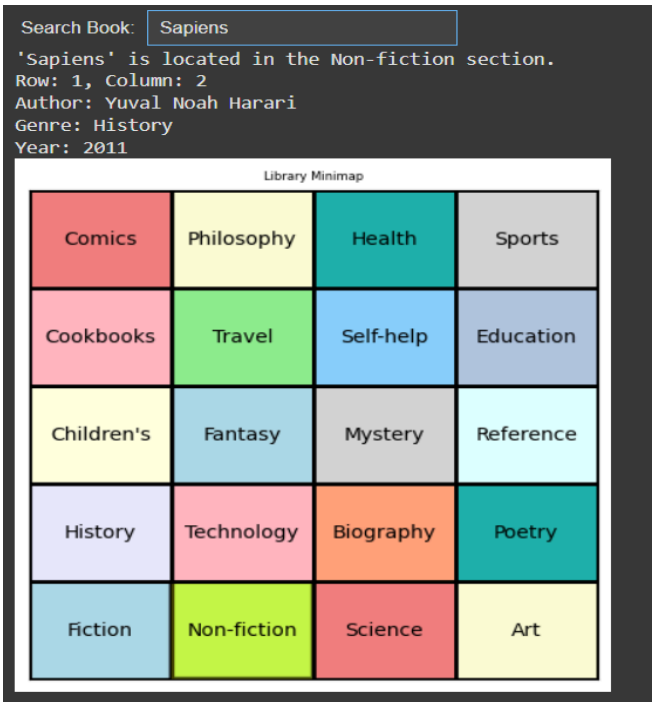


Fig 4 output for the book "Sapiens"

5 . Conclusion

In this research, we explored the organization and accessibility of books in a library setting, leveraging a structured grid-based system to enhance user experience. By mapping book sections to specific locations within the library, we created a more intuitive search and navigation process. Our interactive model, combining visualization with real-world book data, demonstrates how technology can streamline book retrieval, reducing the time spent searching for materials.

The implementation of a library minimap, combined with an interactive book search function, provides an efficient method for users to locate books while also offering a broader understanding of the library's layout. Future enhancements could incorporate machine learning for personalized recommendations, real-time book availability updates, and integration with mobile applications for seamless user access.

Overall, this research highlights the potential of combining data visualization with interactive technologies to improve the efficiency and user-friendliness of library systems. By making book searching more accessible and engaging, we contribute to a more effective and modernized library experience.

6 . Reference

[1] G. F. McKeen, *Islandora: A Drupal/Fedora Repository System*, Islandora Foundation, 2012.

[2] Y. K. Leong, *E-LIS: An International Open Archive Towards Building Open Digital Libraries*, Open Digital

Library Press, 2015.

[3] M. Casey and L. Savastinuk, *Library 2.0: A Guide to Participatory Library Service*, American Library Association, 2007.

[4] Tsinghua University Library, *Smart Talking Robot Xiaotu: AI-Based Library Services*, AI Research Press, 2018.

[5] N. Kazi, *Towards Serving User 2.0: Next-Generation Library Services*, Digital Information Services, 2013.

[6] J. West, *What We Want: An OPAC Manifesto*, Library Technology Reports, 2012.

[7] D. Johnston and A. McDonald, *Data Management in the United States and Canada: Academic Libraries' Contribution*, Springer, 2020.

[8] S. P. Douglas, *Developing a Digital Repository Infrastructure for Sharing Digital Resources in the Teaching Education Community*, Educational Research Press, 2016.

[9] D. Salo, *Retooling Libraries for the Data Challenge*, Library Future Studies, 2017.

[10] K. Shearer and A. Argáez, *Addressing the Research Data Gap: A Review of Novel Services for Libraries*, Research Data Alliance, 2019.

[11] Association of Research Libraries, *New Roles for New Times: Digital Curation for Preservation*, ARL Publications, 2015.

[12] N. Ruest, *The Islandora Web ARChive Solution Pack: Integrating Web Archiving into Digital Libraries*, Islandora Foundation, 2021